

Complete Core Java Mastery Syllabus

© 2026 Ajayeswaran Raja. All Rights Reserved.

Complete Core Java Mastery Syllabus

© 2026 Ajayeswaran Raja. All Rights Reserved.

Program Overview

This syllabus is designed to transform beginners into industry-ready Java developers with strong foundations in Core Java Programming, OOP, Collections, Exception Handling, Multithreading, JDBC, Java 8 Features, and Backend Fundamentals.

Phases Included:

1. Java Fundamentals
2. Control Statements
3. Arrays & Strings
4. Object-Oriented Programming
5. Advanced Java Concepts
6. Collections Framework
7. Multithreading
8. Java 8 Features
9. JDBC
10. Backend Concepts
11. DSA with Java
12. Testing & Debugging
13. Real-Time Projects
14. Interview Preparation

Key Topics:

- JVM, JRE, JDK
- Variables & Data Types
- Operators & Loops
- Arrays & Strings

- Classes & Objects
- Inheritance & Polymorphism
- Exception Handling
- File Handling
- Collections Framework
- Generics
- Threads & Synchronization
- Lambda Expressions
- Streams API
- JDBC CRUD Operations
- REST API Basics
- DSA using Java

Projects:

- Student Management System
- Banking System
- Inventory Management
- E-commerce Backend
- Hospital ERP System
- Real-time Chat Application

Recommended Tools:

- IntelliJ IDEA
- Eclipse
- VS Code
- MySQL
- Git & GitHub
- Maven & Gradle

Career Outcomes:

- Java Developer
- Backend Developer

- Full Stack Developer
- Software Engineer
- Enterprise Application Developer

Final Advice:

Practice coding daily, master OOP concepts deeply, build real-world projects, and solve DSA problems consistently.